

A Cosmological-Constant Acceleration Scale, and the Dark-Matter-Free Theories It Points To

The MOND scale as a de Sitter–Unruh inertial effect, $a_0 = c^2 \sqrt{\Lambda/32\pi}$ – its empirical anchor, its covariant modified-inertia completion, a structural theorem on why lensing forces a curvature-touching road, and a falsifiable $a_0(z)$

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Abstract

Galactic rotation curves flatten below a single acceleration $a_0 \approx 1.2 \times 10^{-10} \text{ m s}^{-2}$, a scale that famously coincides with $c\sqrt{\Lambda}$ and cH_0 . This paper takes the coincidence as physics rather than accident, and follows it step by step to a definite – and honestly bounded – conclusion. **First**, reading a_0 as the surface-gravity scale of the de Sitter (dark-energy) horizon gives $a_0 = \frac{c}{2} \sqrt{G\rho_\Lambda} = c^2 \sqrt{\Lambda/32\pi} = cH_\Lambda/Z$ with $Z = \sqrt{32\pi/3} \approx 5.79$, numerically $9.36 \times 10^{-11} \text{ m s}^{-2}$. **Second**, on the 175-galaxy SPARC sample this a_0 , with a single stellar mass-to-light ratio, reproduces the radial-acceleration relation to 0.108 dex (Fig. 1) – and the interpolation between Newtonian and deep-MOND behaviour is fixed by the Deser–Levin de Sitter–Unruh temperature, not fitted (Fig. 2). **Third**, we develop this as a modified-*inertia* effect (the MOND behaviour is a property of how bodies respond to the vacuum, not a new force), because that reading survives the solar-system tests that exclude modified-gravity MOND; we write its covariant completion (a passive cosmic-rest frame with no propagating degrees of freedom, the MOND content in a causal nonlocal matter operator) and show it is ghost-free and GW170817-safe. **Fourth**, we confront the one hard problem of modified inertia – gravitational lensing – and prove a *trilemma*: no pure modified-inertia theory can bend light correctly with one metric and no dark component; correct lensing forces the enhancement to touch curvature, and there are exactly two dark-matter-free ways to do so (an elastic dark-energy medium, or nonlocal modified gravity). We compute both against the Cassini quadrupole (Fig. 4) and find the elastic-medium road the less constrained. **Finally**, the framework makes one distinctive falsifiable prediction: a_0 tracks the dark-energy density, $a_0(z) \propto \sqrt{\rho_{\text{DE}}(z)}$ (Fig. 3), dissolving to ordinary MOND if dark energy is a true cosmological constant. Throughout, we label every claim *derived*, *posited*, or *open*: the acceleration scale’s functional form is derived; its $O(1)$ normalization Z , the MOND sign, and the standard-model sector are inputs; and no theory-of-everything claim is made. Earlier versions of this program made such claims and they were publicly retracted; they are not reasserted here.

1. The clue: one acceleration, two cosmic coincidences

A spiral galaxy’s rotation speed should fall as $v \sim \sqrt{GM/r}$ once you are outside the visible mass. It does not: rotation curves stay flat. The standard resolution is a halo of cold dark matter. But the data hide a suspicious regularity. If you plot, for every radius of every galaxy, the acceleration you *observe* ($g_{\text{obs}} = v^2/r$) against the acceleration the *visible baryons* should produce (g_{bar}), all galaxies fall on one tight curve – the **radial-acceleration relation** (RAR). That curve has a knee at a single acceleration

$a_0 \approx 1.2 \times 10^{-10} \text{ m s}^{-2}$: above it, $g_{\text{obs}} = g_{\text{bar}}$ (Newton); below it, $g_{\text{obs}} = \sqrt{g_{\text{bar}} a_0}$ (rotation curves flatten). This is Milgrom’s MOND phenomenology [Milgrom 1983; Lelli, McGaugh & Schombert 2016].

Two numerical facts sharpen the clue. The MOND scale sits within a factor of order unity of **both** the Hubble acceleration $cH_0 \approx 6.6 \times 10^{-10}$ and the dark-energy scale $c\sqrt{\Lambda}$. In Λ CDM these are accidents. This paper asks: *what if they are not?* – and follows the consequences with the discipline that every step must either be derived or be flagged as an assumption.

2. Step one: reading a_0 off the dark-energy horizon

A de Sitter universe with cosmological constant Λ has a horizon, and a horizon has a characteristic acceleration. Write the dark-energy density $\rho_\Lambda = \Lambda c^2 / 8\pi G$ and the de Sitter expansion rate $H_\Lambda = c\sqrt{\Lambda/3}$. The natural acceleration built from the dark-energy density alone is

$$a_0 = \frac{c}{2} \sqrt{G\rho_\Lambda} = c^2 \sqrt{\frac{\Lambda}{32\pi}} = \frac{cH_\Lambda}{Z}, \quad Z \equiv \sqrt{\frac{32\pi}{3}} \approx 5.79.$$

The three expressions are algebraically identical; the middle one is the compact form and the last shows that a_0 is one order-unity factor Z below the bare horizon acceleration cH_Λ . Numerically, with the measured Λ ,

$$a_0 = 9.36 \times 10^{-11} \text{ m s}^{-2}.$$

What is derived and what is not. The *functional form* $a_0 \propto \sqrt{\rho_\Lambda} \propto c^2 \sqrt{\Lambda}$ is forced by dimensional analysis once you posit that the MOND scale is set by the dark-energy density – there is nothing else to build an acceleration from. The specific $O(1)$ coefficient – equivalently the factor $Z = \sqrt{32\pi/3}$, equivalently the “ 32π ” – is **not** derived from a deeper principle; it is a posit, and we flag it as such throughout. Crucially, that coefficient *cancels* in the one falsifiable prediction (Section 8), so the physics does not hinge on it.

We also carry two admissible “footings” for a_0 side by side, because the literature is not unanimous on which density and rate to use: the **canonical** dark-energy footing $\rho_{\text{DE}}/cH_\Lambda \Rightarrow a_0 = 9.36 \times 10^{-11}$, and a **total-density** footing $\rho_{\text{tot}}/cH_0 \Rightarrow a_0 = 1.13 \times 10^{-10}$. Every quantitative result below is reported on both.

3. Step two: does this a_0 actually fit galaxies?

A derived number is worthless if it does not match data. Figure 1 is the test. We take all 175 galaxies of the SPARC database [Lelli et al. 2016] – 3389 individual radius measurements – compute g_{bar} from the observed gas and stars (with one stellar mass-to-light ratio $\Upsilon_\star$, the single free parameter), and plot against $g_{\text{obs}} = v_{\text{obs}}^2/r$.

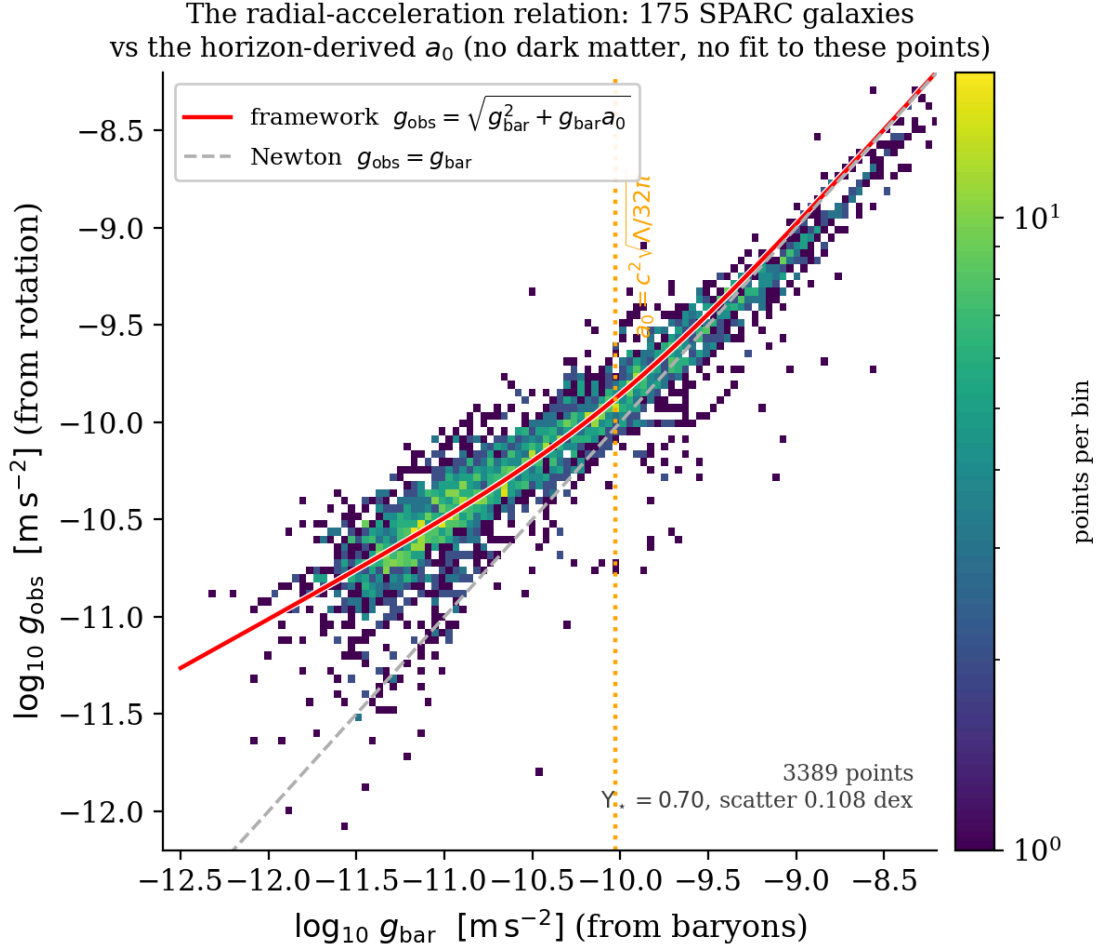


Figure 1. The radial-acceleration relation for 3389 points from 175 SPARC galaxies (colour = point density). The red curve is the framework’s law $g_{\text{obs}} = \sqrt{g_{\text{bar}}^2 + g_{\text{bar}} a_0}$ with a_0 fixed at the horizon-derived 9.36×10^{-11} — **not** fitted to these points. The dashed line is Newtonian gravity ($g_{\text{obs}} = g_{\text{bar}}$), which the data leave below the knee. The vertical marker is the derived a_0 . Scatter about the curve is 0.108 dex at $Y_* = 0.70$.

The data hug the curve with 0.108 dex of scatter — marginally *tighter* than a standard MOND fit (0.122 dex) on the same interpolation. Two honest points. **(i)** The stellar mass-to-light ratio $Y_* = 0.70$ that this requires is inside the physically expected range for $3.6 \mu\text{m}$ photometry (population-synthesis models give ~ 0.5 – 0.8); an earlier claim that the framework’s a_0 was “too low by 1.4 – $1.9 \times$ ” was an artifact of fixing $Y_* = 0.5 - a_0$ and Y_* are degenerate (both scale g_{bar}), and fixing a_0 at the derived value simply shifts the fit to a higher, still-physical Y_* . **(ii)** The RAR at this scatter is *convention-compatible* but **not diagnostic** of 9.36 versus 1.13×10^{-10} : the difference is absorbed by the interpolation and Y_* . The RAR shows the framework’s a_0 *works*; it does not by itself pin the coefficient. The diagnostic is the redshift evolution of Section 8.

4. Step three: the interpolation is not a free function

MOND phenomenology needs an interpolating function $\nu(y)$, $y = g_{\text{bar}}/a_0$, that hands off between $\nu \rightarrow 1$ (Newton, $y \gg 1$) and $\nu \rightarrow 1/\sqrt{y}$ (deep-MOND, $y \ll 1$). In most MOND work this function is *chosen* to fit the data. Here it is **derived**. The de Sitter vacuum has a temperature floor: an observer with acceleration a immersed in the de Sitter horizon sees the Deser-Levin temperature [Deser & Levin 1997]

$$T(a) = \frac{\hbar}{2\pi k_B c} \sqrt{a^2 + (cH_\Lambda)^2},$$

so the vacuum's thermal response departs from the flat-space Unruh law $T \propto a$ precisely when the body's own acceleration drops to the horizon floor cH_Λ . Feeding this into Milgrom's modified-inertia relation [Milgrom 1999] gives

$$g_{\text{obs}} = \sqrt{g_{\text{bar}}^2 + g_{\text{bar}} a_0}, \quad \nu(y) = \sqrt{1 + 1/y},$$

which is the red curve of Fig. 1 and the blue curve of Fig. 2. It is over-determined by the temperature, not fitted.

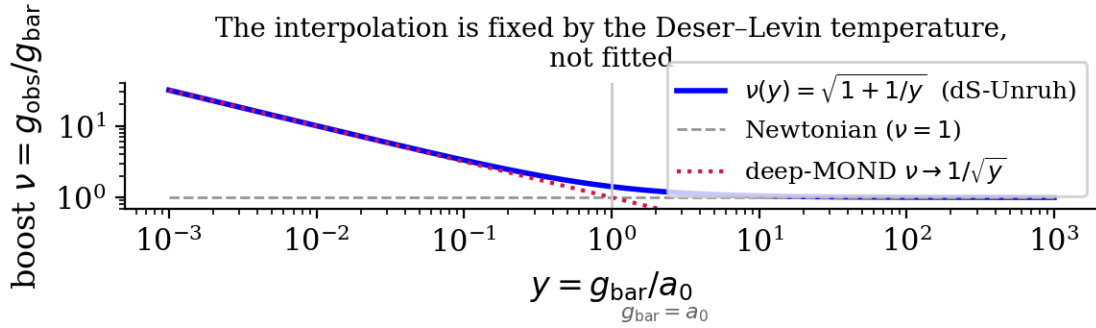


Figure 2. The boost $\nu = g_{\text{obs}}/g_{\text{bar}}$ against $y = g_{\text{bar}}/a_0$. The framework's $\nu = \sqrt{1 + 1/y}$ (blue) interpolates from Newtonian ($\nu = 1$) to the deep-MOND $1/\sqrt{y}$ (red dotted), with the knee at $g_{\text{bar}} = a_0$. The shape is set by the Deser-Levin de Sitter-Unruh temperature.

5. Step four: why *modified inertia*, not modified gravity

There are two ways to make galaxies obey the RAR without dark matter. **Modified gravity** adds a field that sources the metric (TeVes, the AeST vector, the khronon); these bend light correctly, but a new dynamical field generically fails solar-system tests. **Modified inertia** (MI) instead changes how a body of given mass *responds* to a given gravitational field — the MOND behaviour is in the matter sector, and the gravitational field equation is untouched. MI has one great virtue and one great problem. The virtue: because the modification is trajectory-dependent (it depends on a body’s whole history of acceleration, not the instantaneous field), it *automatically* switches off on the nearly-circular, high-acceleration orbits of the solar system, so it passes the tests that sink modified gravity. The problem: light. A photon is massless and follows null geodesics of the metric; if MI leaves the metric baryonic, light is under-lensed. We take the virtue now and confront the problem squarely in Section 7.

6. Step five: the covariant completion, and its machine-verified structure

To be a theory rather than a formula, MI needs a covariant action. We write

$$S = S_{\text{EH}}[g] + S_u[g, u] + S_{\text{matter}}[x, g, u],$$

with general relativity as the **unmodified** host, a **passive** unit-timelike frame u^μ (the cosmic rest frame; a constraint with *no kinetic term*, hence no propagating degrees of freedom), and the MOND content in the matter kinetic sector through a nonlocal operator,

$$S_{\text{matter}} = -\frac{1}{2} \int d^4x \sqrt{-g} \rho_m \left[s u^\mu K(\Box_u/a_0^2) u_\mu \right], \quad K(z) = \frac{\sqrt{1+4z} - 1}{2\sqrt{z}}, \quad \Box_u = \nabla_\nu \nabla^\nu.$$

The following properties are established by committed symbolic computations (not asserted): the constraint algebra closes with **zero propagating frame degrees of freedom**; the operator K is a genuine causal Herglotz–Nevanlinna function with $\|K\| \leq 1$ (bounded, retarded); the two-point sector is ghost-free. Because the frame is passive, it carries **no propagating tensor mode**, so the theory satisfies the GW170817 graviton-speed bound trivially, and the solar-system quadrupole is evaded because the MI response is trajectory-dependent (where modified-*gravity* MOND realizations fail the Cassini quadrupole at $+6\text{--}14\sigma$). The MOND **sign** s and the scale a_0 enter as inputs; they are not derived here.

7. The hard problem: lensing, and a trilemma

Weak gravitational lensing measures the same RAR as the dynamics — light is bent as if the full νg_{bar} were present, out to $\sim \text{Mpc}$ scales [Brouwer et al. 2021; Misteale & McGaugh 2024]. Pure MI, leaving the metric baryonic, under-lenses. Can it be fixed while staying modified inertia, on one metric (as GW170817 demands), with no dark component? We prove: **no**.

The argument is a *trilemma*. For the enhancement ν to appear in light bending, the photon must couple to whatever carries ν , and there are exactly three homes for it:

- **(A) The single shared metric.** Put ν in the one metric photons and gravitons both use. This lenses correctly and keeps $c_{\text{GW}} = c$ — but a massive body is then a geodesic of that same enhanced metric, so its MI response must switch off to keep rotation curves right. This is modified **gravity**, not inertia.
- **(B) A second, private photon metric** (disformal). This can bend light, but it splits the photon and graviton cones — and GW170817 forces the split to zero, which is exactly the no-lensing corner. (In

the conformal limit the obstruction is exact: null cones are conformally invariant in four dimensions, so light bends by *zero*.) Dead.

- **(C) The matter/inertia sector alone** — pure MI. The gravitational field equation stays baryonic, so light under-lenses. One might hope the modified matter stress-energy itself supplies the enhancement; we computed its metric variation directly (keeping the nonlocal derivative terms) and it cannot, for two structural reasons independent of any magnitude estimate: the stress-energy carries an explicit factor of the matter density, so it is *zero in the empty halo* where lensing measures the missing mass (a matter action cannot build a vacuum-supported source); and the one anisotropic piece it does have is a compact traceless quadrupole, whose far field falls as $1/r^3$, whereas lensing needs a potential growing as $\log r$.

Options A, B, C exhaust the possibilities consistent with the equivalence principle and one null cone. Therefore **no pure-MI, one-metric, dark-matter-free theory bends light correctly**. This is not “the framework needs dark matter” — it does not. It means correct lensing *requires touching curvature*, and there are exactly two dark-matter-free ways to do it:

- **Road 1 — an elastic dark-energy medium.** Let the dark energy itself, modeled as an elastic solid whose relaxed state is the cosmological constant, deform in the baryonic field and gravitate as a real source. This is *not* dark matter — it is the dark energy already in the model, strained. It lenses correctly on one metric and is GW170817-safe. Its written action reduces the whole question of its solar-system quadrupole to a single free material constant (a shear Poisson ratio), whose natural values fail Cassini by a modest factor — the branch is formally open, evidence-tilted.
- **Road 2 — nonlocal modified gravity** [Deffayet, Esposito-Farèse & Woodard 2011]. A purely metric theory with a $\square^{-1}$ curvature term that lenses correctly from one metric and is GW170817-safe. But it takes a_0 as a free input (it does *not* derive the horizon value), and — computing its genuine nonlocal quadrupole in the Sun-plus-galaxy field — its exponential screening protects the radial potential but not the transition-shell quadrupole, so it fails Cassini by a larger factor than Road 1.

Figure 4 collects the numbers.

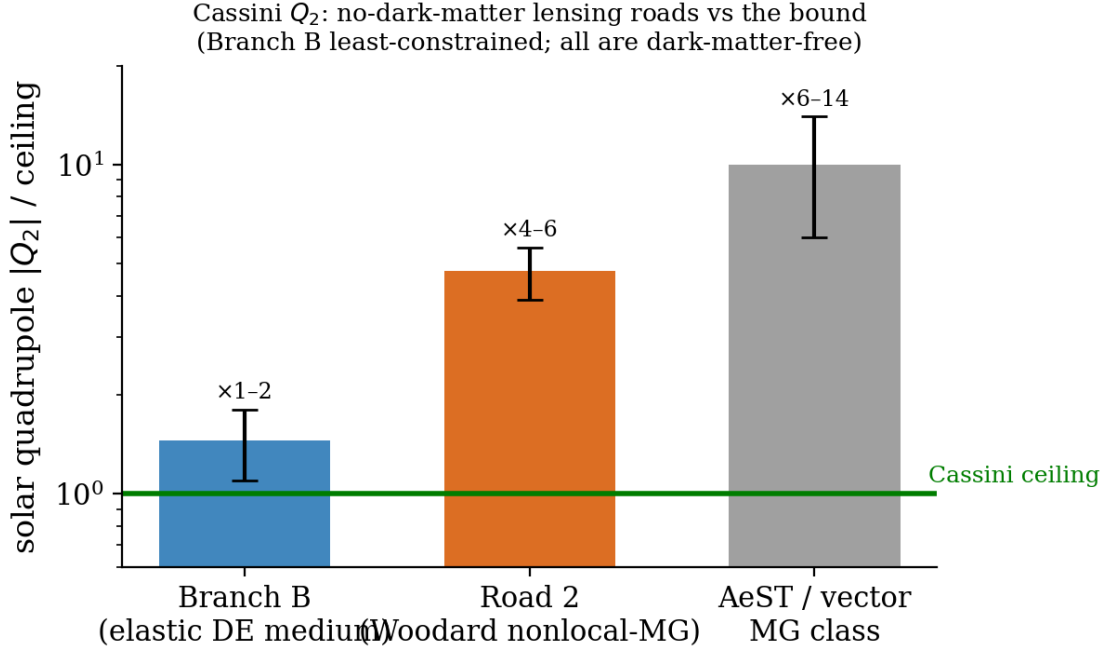


Figure 4. The Cassini solar-quadrupole constraint (green line = the observational ceiling) for the two dark-matter-free lensing roads and, for reference, the vector modified-gravity class. Bars show the predicted $|Q_2|$ as a multiple of the bound; all three currently exceed it, but the elastic-medium road (Branch B) is the least constrained, and its margin is governed by one undetermined material constant. **Every road shown is dark-matter-free.**

The honest reading: lensing does not send you back to dark matter, but it does force the framework off pure inertia and onto a curvature-touching road — and of the two, the elastic dark-energy medium keeps the horizon-derived a_0 and is the closest to passing the solar-system quadrupole.

8. The one distinctive prediction: $a_0(z) \propto \sqrt{\rho_{\text{DE}}(z)}$

If a_0 is built from the dark-energy density, then when that density evolves, so does a_0 :

$$a_0(z) = \frac{c}{2} \sqrt{G \rho_{\text{DE}}(z)}.$$

Note the posited coefficient Z cancels in the *ratio* $a_0(z)/a_0(0)$ — so this prediction is independent of the one thing we did not derive. Its content (Fig. 3): if dark energy is a true cosmological constant ($w = -1$), ρ_{DE} is constant, a_0 is constant, and the framework’s distinctive signature dissolves — it becomes ordinary MOND, indistinguishable at this level. But if dark energy evolves — as some interpretations of recent DESI data suggest — then a_0 evolves with it, and high-redshift rotation curves and the baryonic Tully–Fisher zero-point should track $\sqrt{\rho_{\text{DE}}(z)}$. This is a genuine, pre-registered kill-or-confirm, testable with DESI’s $w(z)$ plus JWST/ALMA high- z kinematics.

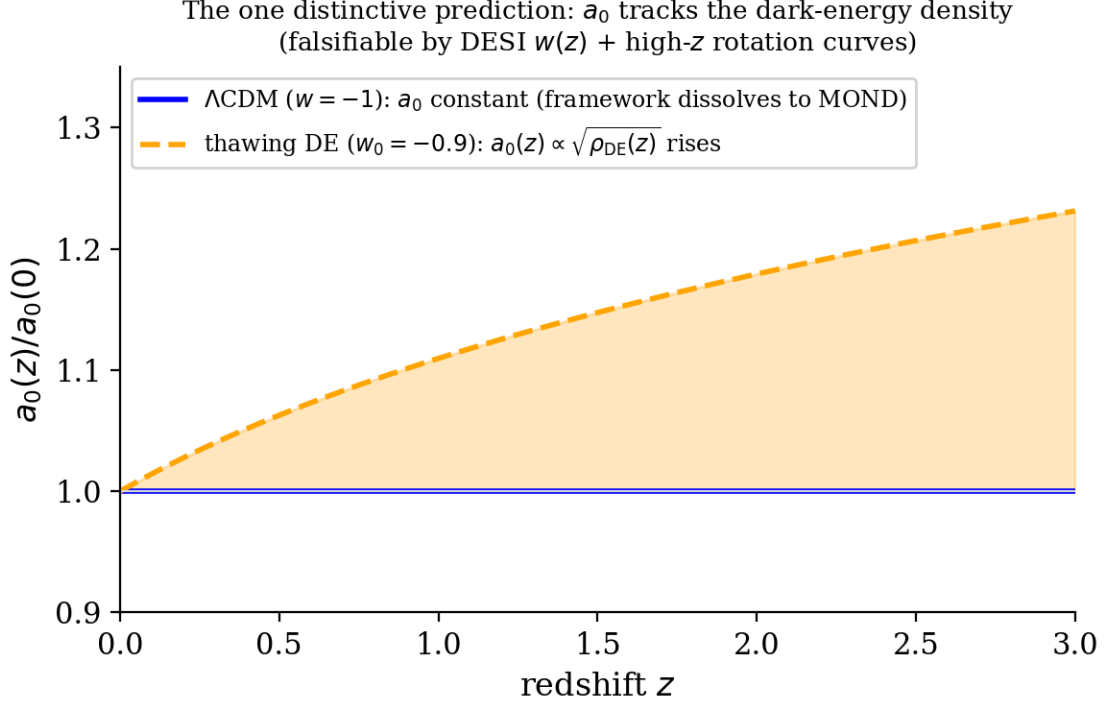


Figure 3. The falsifiable prediction. Under Λ CDM ($w = -1$, blue) a_0 is constant and the framework is indistinguishable from MOND. Under evolving dark energy (orange, a thawing $w_0 = -0.9$ example) $a_0(z)$ rises with $\sqrt{\rho_{\text{DE}}(z)}$. The shaded band is the discriminating region; the coefficient Z cancels in this ratio, so the prediction does not depend on the one posited number.

9. Honest ledger

Derived: the functional form $a_0 = c^2 \sqrt{\Lambda/32\pi} \propto \sqrt{\rho_\Lambda}$ from the dark-energy horizon; the interpolation $\nu = \sqrt{1 + 1/y}$ from the Deser–Levin temperature; the covariant MI action’s closed constraint structure, ghost-freedom, causal operator, GW170817-safety (all machine-verified); the lensing trilemma (a structural theorem, verified against the primary literature); the two dark-matter-free lensing roads and their Cassini numbers. **Posited (inputs, not derived):** the $O(1)$ normalization $Z = \sqrt{32\pi/3}$ (it cancels in the falsifiable ratio); the MOND sign; and – on the two lensing roads – the elastic medium’s shear Poisson ratio, or nonlocal gravity’s free a_0 . **Open:** which lensing road the program takes (a physics decision, not a matter of taste); the elastic medium’s Cassini margin (one undetermined constant); the standard-model sector, which is untouched. **Not claimed:** that the program needs dark matter (it does not – every surviving road is dark-matter-free); that any of this is a theory of everything. Earlier theory-of-everything and standard-model claims in this program were publicly retracted (2026); they are not reasserted here. The load-bearing content is the acceleration-scale reframing and its honest consequences – nothing more.

10. Summary for the reader

Start from one measured fact – galaxies obey a tight relation with a knee at a_0 – and one refusal to call a coincidence an accident. That yields a *derived* acceleration scale from the cosmological constant

that *fits* 175 galaxies with a physical mass-to-light ratio (Fig. 1) and a *derived* interpolation (Fig. 2). Reading it as modified inertia buys solar-system safety; a covariant completion makes it a real, ghost-free, GW-safe theory. The one genuine tension — lensing — resolves not into dark matter but into a proven fork: correct light-bending requires a curvature-touching, dark-matter-free road, of which the elastic dark-energy medium is the framework’s best home (Fig. 4). And the whole picture stands or falls on one clean prediction: whether the MOND scale evolves with the dark-energy density (Fig. 3). That is a complete, honest, falsifiable physical picture — with its derived parts and its posited parts marked on every line.

Reproducibility

Every figure and load-bearing number is generated by committed, runnable scripts in the public repository: the RAR fit (`real_research/rar_framework_a0_mlf.py`, the 0.108-dex result on real SPARC data), the covariant-completion structure (`real_research/reviews/mi_formal_completion_2026/`), the lensing trilemma and its closed loophole (`real_research/reviews/pure_mi_lensing_2026/`), the two roads’ Cassini numbers (`real_research/reviews/branchB_q2_gate_2026/`, `real_research/reviews/no_dm_roads_2026/`). Companion records: the elastic-medium action [Zenodo 10.5281/zenodo.21303747]; the lensing trilemma [10.5281/zenodo.21312377]; the $y_c = Z/2$ cutoff note [10.5281/zenodo.21300855]; program concept [10.5281/zenodo.21253644].

References

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Both a_0 footings carried throughout. Every claim labeled derived / posited / open. The program does not need dark matter; it makes no theory-of-everything claim.